

Tiger Lu

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APPOINTMENTS	Center for Computational Astrophysics , Flatiron Institute, New York, NY Flatiron Research Fellow	Sep 2025 – Present
	Yale University , New Haven, CT Graduate Research Assistant	Aug 2020 – May 2025
	University of California, Berkeley , Berkeley, CA Summer Undergraduate Researcher	May 2019 – Aug 2019
	Carnegie Observatories , Pasadena, CA Summer Undergraduate Researcher	May 2018 – Aug 2018
EDUCATION	Yale University , New Haven, Connecticut, USA	
	▪ Ph.D. in Astronomy with Distinction • Thesis: Planets are not Points: the Profound Effect of Planetary Structure on Exoplanet System Architectures • Advisers: Prof. Gregory Laughlin, Prof. Malena Rice	Aug 2020 – May 2025
	▪ M.S., M.Phil. in Astronomy	Aug 2020 – Aug 2022
	California Institute of Technology , Pasadena, California, USA	
	▪ B.S. in Astrophysics ▪ Minor in Computer Science	Aug 2016 – Jun 2020
PUBLICATIONS	155 citations. h -index = 8. <i>Italics:</i> Student directly supervised by T. Lu	
	FIRST-AUTHOR	
	[7] Lu, T. , Hernandez, D.M., Rice, M. and Su, Y. in review.	
	[6] Lu, T. , Tajer, H., Hernandez, D.M., Rein, H., <i>Liu</i> , Y. and Rice, M., 2025. Collisional Fragmentation Support in TRACE . <i>RNAAS</i> 9 , 5	
	[5] Lu, T. , Li, G., Cassese, B. and Lin, D.N.C., 2025. The Dynamical History of HIP-41378 f - Oblique Exorings Masquerading as a Puffy Planet . <i>ApJ</i> 980 , 39	
	[4] Lu, T. , An, Q., Li, G., Millholland, S.C., Rice, M., Brandt, G.M. and Brandt, T., 2025. Planet-Planet Scattering and ZLK Migration – the Dynamical History of HAT-P-11 . <i>ApJ</i> 979 , 218	
	[3] Lu, T. , Hernandez, D.M. and Rein, H., 2024. TRACE: a Time-Reversible Algorithm for Close Encounters . <i>MNRAS</i> 533 , 3708	
	[2] Lu, T. , Rein, H., Tamayo, D., Hadden, S., Mardling, R., Millholland, S.C. and Laughlin, G., 2023. Self-Consistent Spin, Tidal and Dynamical Equations of Motion in the REBOUNDx Framework . <i>ApJ</i> 948 , 41	
	[1] Lu, T. and Laughlin, G., 2022. Tilting Uranus via Secular Spin-Orbit Resonance with Planet 9 . <i>PSJ</i> 3 , 221	
	STUDENT-LEAD	
	[3] <i>Im</i> , H., Lu, T. , Rice, M., Tran, Q., Li, G. and Naoz, S. in review.	
	[2] <i>Liu</i> , Y., Lu, T. and Rice, M., 2025. Double Hot Jupiter Formation through Mirrored ZLK Migration in Binary Star Systems: The Case of WASP-94 . <i>ApJL</i> 997 , L41	
	[1] <i>Liu</i> , Y., Lu, T. and Rice, M., 2025. The Formation of Double Hot Jupiter Systems Through ZLK Migration . <i>ApJ</i> 986 , 103	
	SIGNIFICANT CONTRIBUTION	
	[5] Yahalomi, D.A., Lu, T. , Armitage, P., Bedell, M., Casey, A.R., Price-Whelan, A., and Rice, M., 2026. The Astrometric Resoeccentric Degeneracy: Eccentric Single Planets Mimic 2:1 Resonant Planet Pairs in Astrometry . <i>ApJL</i> in press	

- [4] Cassese, B., Rice, M. and **Lu, T.**, 2025. [A High-Precision, Differentiable Code for Solar System Ephemerides](#). *PSJ* **6**, 252
- [3] An, Q., **Lu, T.**, Brandt, G.M., Brandt, T. and Li, G., 2025. [Significant Mutual Inclinations Between the Stellar Spin and the Orbits of Both Planets in the HAT-P-11 System](#). *AJ* **169**, 22
- [2] Cassese, B., Vega, J., **Lu, T.**, Rice, M., Poddar, A. and Kipping, D., 2024. [squishyplanet: Modeling transits of non-spherical planets in JAX](#). *JOSS* **9**, 100
- [1] Wahl, S., Thorngren, D., **Lu, T.** and Militzer, B., 2021. [Tidal Response and Shape of Hot Jupiters](#) *ApJ* **921**, 105

OTHER CO-AUTHOR

- [3] Tajer, H., Wang, J., Childs, C., et al. (inc. **Lu, T.**) 2026. [Origins of Mercury’s Big Heart of Iron: Exploring Pathways to Form High Core Mass Fraction \(CMF\) Planets via N-body Simulations](#). *ApJ* **998**, 46
- [2] Bardalez Gagliuffi, D.†, Balmer, W.†, Pueyo, L., et al. (inc. **Lu, T.**) 2025. [JWST Coronagraphic Images of 14 Her c: a Cold Giant Planet in a Dynamically Hot, Multi-planet System](#). *ApJL* **988**, L18
- [1] Wang, N., Lu, L.Y., Liu, H.G., et al. (inc. **Lu, T.**) 2025. [Development of Gravity Theories in the View of TRAPPIST-1e](#). *RAA* **25**, 015003

WHITEPAPERS

- [1] Levine, W. G., Gerbig, K., Loudon, E., **Lu, T.**, Hsieh, C., O’Connor, C., Li, R. and Dong, J., 2024. [Emerging Researchers in Exoplanetary Science \(ERES\): Lessons Learned in Conference Organization for Early-Career Researchers](#). *Bulletin of the AAS* **56**, 2024i004

†: Joint first author

AWARDS & FELLOWSHIPS

▪ DDA/AAS Raynor L. Duncombe Student Research Prize	2025
▪ Flatiron Research Fellowship	2025
▪ Flipped Science Fair Best Poster Explanation	2024
▪ AAS International Travel Grant	2024
▪ Graduate Student Assembly Conference Travel Fellowship	2024
▪ National Science Foundation GRFP (<i>Honorable Mention</i>)	2022
▪ Joan and Arnold Seidel Griffith Observer Science Writing Contest (<i>Second Place</i>)	2022
▪ Alan Porter Memorial Foundation Fellow	2019
▪ Summer Undergraduate Research Fellow	2018

ORAL PRESENTATIONS & TALKS

▪ <i>Astronomy & Planetary Science Colloquium, NAU* (Flagstaff, AZ)</i>	Feb 2026
▪ <i>Astronomy Seminar, AMNH* (Flagstaff, AZ)</i>	Feb 2026
▪ International Conference on Exoplanets and Planet Formation (<i>Shanghai, China</i>)	Dec 2025
▪ Powell Group Meeting, University of Chicago (<i>Chicago, IL</i>)	Oct 2025
▪ 246th AAS Meeting Dissertation Talk (<i>Anchorage, AK</i>)	Jun 2025
▪ Boston Area Planetary Science Meeting, MIT (<i>Cambridge, MA</i>)	May 2025
▪ New York Area Exoplanets Meeting II, CCA (<i>New York City, NY</i>)	May 2025
▪ 56th DDA Meeting, Georgia Institute of Technology (<i>Atlanta, GA</i>)	May 2025
▪ ML in Planetary Dynamics Workshop*, CCA (<i>New York City, NY</i>)	Apr 2025
▪ Tea Talk, Indiana University (<i>Bloomington, IN</i>)	Feb 2025
▪ ITC Luncheon, Harvard University (<i>Cambridge, MA</i>)	Jan 2025
▪ Exoplanet Pizza Lunch, Harvard University (<i>Cambridge, MA</i>)	Jan 2025
▪ Exoplanet Seminar, Princeton University (<i>Princeton, NJ</i>)	Nov 2024
▪ Planetary Lunch Seminar, Massachusetts Institute of Technology (<i>Cambridge, MA</i>)	Nov 2024
▪ Astronomy Seminar, Columbia University (<i>New York City, NY</i>)	Oct 2024
▪ Astronomy Seminar, Stony Brook University (<i>Stony Brook, NY</i>)	Oct 2024
▪ CRA Seminar, Georgia Institute of Technology (<i>Atlanta, GA</i>)	Oct 2024
▪ Planetary Science Seminar, California Institute of Technology (<i>Pasadena, CA</i>)	Oct 2024
▪ Lunch Seminar, UCLA (<i>Los Angeles, CA</i>)	Sep 2024
▪ Exoplanet Group Meeting, UCSB (<i>Santa Barbara, CA</i>)	Sep 2024
▪ FLASH Talk, UCSC (<i>Santa Cruz, CA</i>)	Sep 2024
▪ Star and Planet Lunch, University of Hawai’i (<i>Honolulu, HI</i>)	Aug 2024

- CT Exoplanet Picnic, Wesleyan College (*Wesleyan, CT*) Jul 2024
- Emerging Researchers in Exoplanet Science IX, Cornell University (*Ithaca, NY*) Jul 2024
- REBOUND 24 Meeting* (*Virtual*) Jul 2024
Arbitrary ODE Workshop
- REBOUND 24 Meeting* (*Virtual*) Jul 2024
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- New York Area Exoplanets Meeting I, CCA (*New York City, NY*) May 2024
- 55th DDA Meeting, University of Toronto (*Toronto, Canada*) May 2024
- Makino Lab Seminar, Kobe University (*Kobe, Japan*) Jul 2023
- Exoplanet Seminar, Nanjing University (*Nanjing, China*) Jul 2023
- 54th DDA Meeting, Michigan State University (*East Lansing, MI*) May 2023
- Exoplanet Seminar, Princeton University (*Princeton, NJ*) Mar 2023
- Astronomy Seminar, Monash University (*Monash, Australia*) Feb 2023
- 240th AAS Meeting (*Pasadena, CA*) Jun 2022

*Invited Talk

Italics: Upcoming

POSTER PRESENTATIONS

- Emerging Researchers in Exoplanet Science, Princeton University (*Princeton, NJ*) Jun 2025
- Exoplanets V Meeting (*Leiden, The Netherlands*) Jun 2024
- Extreme Solar Systems (*Christchurch, New Zealand*) Jun 2024
- 2022 AGU Meeting (*Chicago, IL*) Dec 2022
- Exoplanets IV Meeting (*Las Vegas, NV*) May 2022
- 2019 AGU Meeting (*San Francisco, CA*) Dec 2019
- 233rd AAS Meeting (*Seattle, Washington*) Jan 2019

TEACHING

- ASTR 210: Stars and Their Evolution** New Haven, CT, USA
Yale University Teaching Fellow Spring 2023
- ASTR 160: Frontiers and Controversies in Astrophysics** New Haven, CT, USA
Yale University Teaching Fellow Fall 2022
- ASTR 160: Frontiers and Controversies in Astrophysics** New Haven, CT, USA
Yale University Teaching Fellow Spring 2021
- ASTR 155: Introduction to Astronomical Observing** New Haven, CT, USA
Yale University Teaching Fellow Fall 2021
- Dean's Tutor** Pasadena, CA, USA
Tutoring Caltech students in Physics, Astronomy and Computer Science 2017-2020

OUTREACH

- Warrior-Scholar Program** New Haven, CT, USA
Research Project Leader Summer 2023
One week intensive boot camp preparing military veterans for undergraduate degrees
- Yale OpenLabs**
Local outreach organization dedicated to educating middle school children on STEM topics. Receptient of the Seton Elm-Ivy Award.
 - Public Relations Officer 2023
 - Regular Outreach Speaker 2022 – 2025
 - Science Cafe 2024 Speaker: *What if 'Cotton Candy' Planet are Actually Ring Pops?*
 - Flipped Science Fair Best Poster Explanation: *Probing the Shapes of Planets*
 - Science Cafe 2022 Speaker: [How We Discover Exoplanets](#)
- Astronomy on Tap**
New Haven branch of outreach program aimed at engaging local community with astronomy research
 - Social Media Director 2022
- Science in the News**
Local outreach organization dedicated to educating the greater community on STEM topics.
 - Outreach Speaker 2021
 - [Hidden Things: Pulling Back the Veil on the Universe](#)

RESEARCH MENTORING	Haedam Im* Dorrit Hoffleit Undergraduate Research Fellow Project: <i>Hunting for a Hidden Planet in the GJ 436 System</i>	May 2025 – Massachusetts Institute of Technology
	Yurou (Nina) Liu* Alexander P. Hixon Summer Undergraduate Research Fellow Project: <i>The Formation of Double Hot Jupiter Systems Through ZLK Migration</i>	Jun 2023 – Yale University
	Eritas Yang Graduate Student	Oct 2025 – Princeton University
	Vighnesh Nagpal Graduate Student	Aug 2025 – University of Chicago
	Haniyeh Tajer Graduate Student	May 2025 – Ohio State University
	Jeremiah Reynoso Summer Undergraduate Research Fellow *Primary Research Advisor	Jun 2023 – Oct 2023 Morehouse College
INFORMAL MENTORING	Chris Santiago Astro Sibb	2022 – 2024 Yale University
	Kylyn Smith Astro Sibb	2023 – 2024 Yale University
	Sally Jiang (current Ph.D. Candidate, Columbia University) Astro Sibb	2022 – 2023 Yale University
TELESCOPE PROPOSALS (CO-I)	Keck Observatory (KPF) – 0.5 nights <i>Directly Probing the Migration History of WASP-53 with Precise Orbital Constraints</i>	Yale 2026A
	Keck Observatory (KPF) – 2 nights <i>The Search for Double Hot Jupiter Systems</i>	Yale 2024B
	Keck Observatory (HIRES) – 2 nights <i>Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 2)</i>	Yale 2023A
	Keck Observatory (HIRES) – 2 nights <i>Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 1)</i>	Yale 2022B
PROFESSIONAL SERVICE	Science Organizing Committee – 57th DDA Meeting <i>Division of Dynamical Astronomy, University of Chicago</i>	June 2026
	Science Organizing Committee Chair – DSPPS <i>A Dynamical Synthesis of Planetary Systems, New Haven</i>	Jan 2026
	Organizing Committee – ERES X <i>Emerging Researchers in Exoplanet Science Symposium, Princeton University</i>	July 2025
	Science Organizing Committee – ERES VIII <i>Emerging Researchers in Exoplanet Science Symposium, Yale University</i>	June 2023
	Reviewer <i>NASA Solar System Workings Panel, A&A, AJ, ApJ, MNRAS</i>	2024 – Present
	Main Co-Organizer – Yale Exoplanets and Stars Seminar <i>Weekly Seminar Series</i>	2023 – 2025
	Committee Member – ACDC <i>Yale University Astronomy Climate and Diversity Committee</i>	2022 – 2025
MEDIA	Yale Daily News New study offers a double dose of ‘hot Jupiters’	Jun 2025
	Popular Science What if ‘cotton candy’ planets are actually Ring Pops?	Oct 2024

COMMUNITY
ENGAGEMENT

New Haven Road Runners – *New Haven, CT, USA*

Local USATF sanctioned running club

- Public Relations Officer
- Racing Team Director / Elite Athlete Coordinator

Jul 2023 – Jul 2024

Jul 2022 – Jun 2025

NCAA Athlete – *Pasadena, CA, USA*

2016 – 2020

NCAA Division III Cross Country/Track & Field at Caltech

Avery House – *Pasadena, CA, USA*

Student Government

- Athletics Manager
- Underclassmen Counselor

2017 – 2018

2017 – 2020

[CV compiled on 2026-02-17]